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on

“ParentalGuard: A Predictive Analytics Framework for Maternal Health Surveillance”

Submitted in partial fulfillment of the requirement for the award of the degree of

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Computer Science and Engineering

Submitted By

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CERTIFICATE

Certified that the Project Entitled **"PARENTALGUARD: A PREDICTIVE ANALYTICS FRAMEWORK FOR MATERNAL HEALTH SURVEILLANCE"** carried out by **VISHWANATH M KARADIGUDD**, bearing **USN 2JH22CS125,** bonafide student of Jain College of Engineering and Technology, is in partial fulfillment for the award of the BACHELOR OF ENGINEERING in Computer Science and Engineering from **Visvesvaraya Technological University, Belagavi** during the year 2025-2026. It is certified that all the corrections/suggestions indicated for Internal Assessment have been incorporated in the report submitted to the department library. The Project report has been approved as it satisfies the academic requirements in respect of the Project prescribed for the said Degree.

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DECLARATION

I, **VISHWANATH M KARADIGUDD**, bearing **USN 2JH22CS125** students of Seventh Semester B.E, Department of Computer Science and Engineering, Jain College of Engineering and Technology, Sai Nagar, Hubballi, declare that the Project Work entitled "**PARENTALGUARD: A PREDICTIVE ANALYTICS FRAMEWORK FOR MATERNAL HEALTH SURVEILLANCE"** has been carried out by us and submitted in partial fulfillment of the course requirements for the award of degree in **Bachelor of Engineering** in **Computer Science and Engineering** from **Visvesvaraya Technological University, Belagavi** during the academic year **2025-2026**. The matter embodied in this report has not been submitted to any other university or institution for the award of any other degree.

2JH22CS125 VISHWANATH M KARADIGUDD

Place: Hubballi
Date:

ACKNOWLEDGEMENT

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2JH22CS125 VISHWANATH M KARADIGUDD

ABSTRACT

Maternal health is considered one of the significant concerns in public health since it deals with the condition of women during pregnancy and childbirth. Complications at these stages may prove fatal to both the mother and child. thus, early detection and follow-up are of utmost importance. The following paper proposes an intelligent maternal health risk prediction system using machine learning algorithms that can grade risks during the maternal period. The proposed system allows real-time maternal health monitoring by acquiring physiological parameters like heart rate, blood pressure, blood glucose, and body temperature remotely and non-invasively.

This proposed model will be more helpful for pregnant women living in rural or underserved areas where health facilities may not be readily available. Several machine learning models have been considered, such as XGBoost. however, the Random Forest Classifier produces the best performance with an accuracy of 93.14%. EDA identifies the most influential factors in health, while trimester-wise visualizations clearly show maternal risk patterns. Additionally, the system will maintain the privacy of data by securely storing data in an encrypted form and allowing user-level access to data. This project will help in data-driven maternal care, ensure timely medical interventions, and contribute to the reduction of maternal and infant mortality rates.

Keywords: Maternal Mortality Rate (MMR), Remote health monitoring, Data preprocessing, Machine learning, Feature engineering, XGBoost, Explainable AI (XAI), Predictive analytics, MySQL database, Physiological parameters, Blood pressure, Blood sugar level, Body temperature, Heart rate, Preeclampsia, Gestational diabetes, Predictive modelling.

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Chapter 1

INTRODUCTION

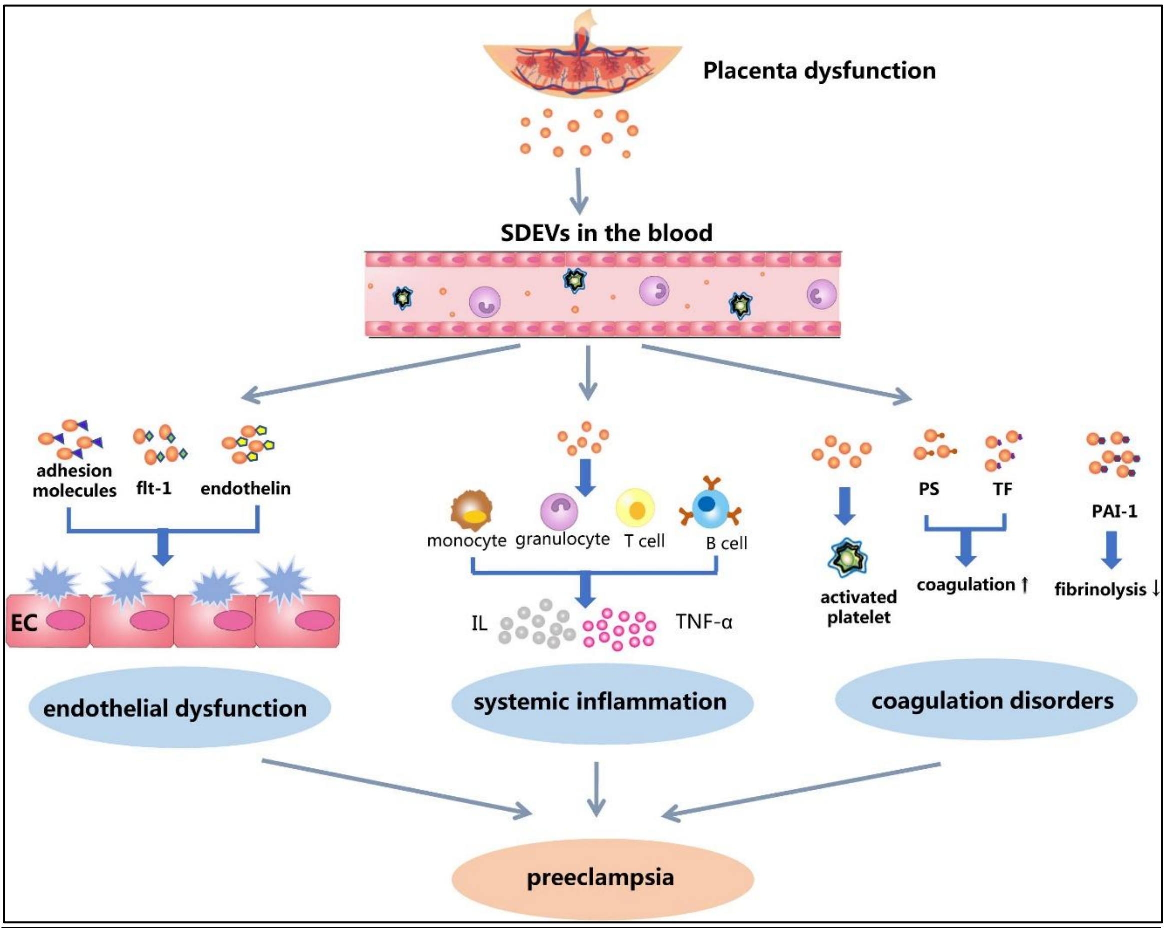
Maternal health is a critical aspect of public health particularly in rural areas where access to healthcare services may be limited. The mortality rate among pregnant women remains a significant concern globally, with various risk factors contributing to adverse outcomes. Many pregnant women face life-threatening complications during pregnancy and post-pregnancy due to a lack of sufficient information about maternal health care. This knowledge gap can lead to preventable deaths among pregnant women and neonates, highlighting the critical need for comprehensive monitoring and early intervention strategies.

Maternal Health Risk (MHR) refers to potential health problems arising during pregnancy, childbirth and postpartum. According to the WHO, there are around 280,000 fatalities of women due to pregnancy complications which means a woman dies approximately every two minutes (WHO, 2023). The Maternal Mortality Rate (MMR) in India was 97 per 100,000 live births in 2018-20. In 2020, India recorded 24,000 maternal deaths making it the second-highest globally. and in Karnataka 217 maternal death cases were registered between August to November in 2024(on average 50 deaths per month). The study was conducted at Karnataka Institute of Medical Sciences, Hubli (In between October 2010 to March 2011), which caters to 250 PHC's/CHC's and is a major referral centre for 4 districts with an average of 800-1000 deliveries per month, followed by the similar reports from Hubballi, where KIMS Hospital confirmed the deaths of 33 pregnant women and 148 infants since January (In 2025). During the study period, among the 40 maternal deaths, 34 deaths (85%) occurred due to direct obstetric causes and 6 deaths (15%) due to indirect causes.

The various factors increase the mortality rate of maternal women and childbirth, including the shortage of doctors and nurses and the localization, time, and distance (Redondi et al., 2013). According to WHO's report in 2020, around 800 women die daily due to poor resources and care (Castillejo et al., 2013). Despite recent technological advances, the rate of maternal death is decreasing, making it difficult to ensure both the mother's and child's safety during pregnancy. Pregnancy-related risks can be reduced in this scenario by anticipating complications and taking precautions.

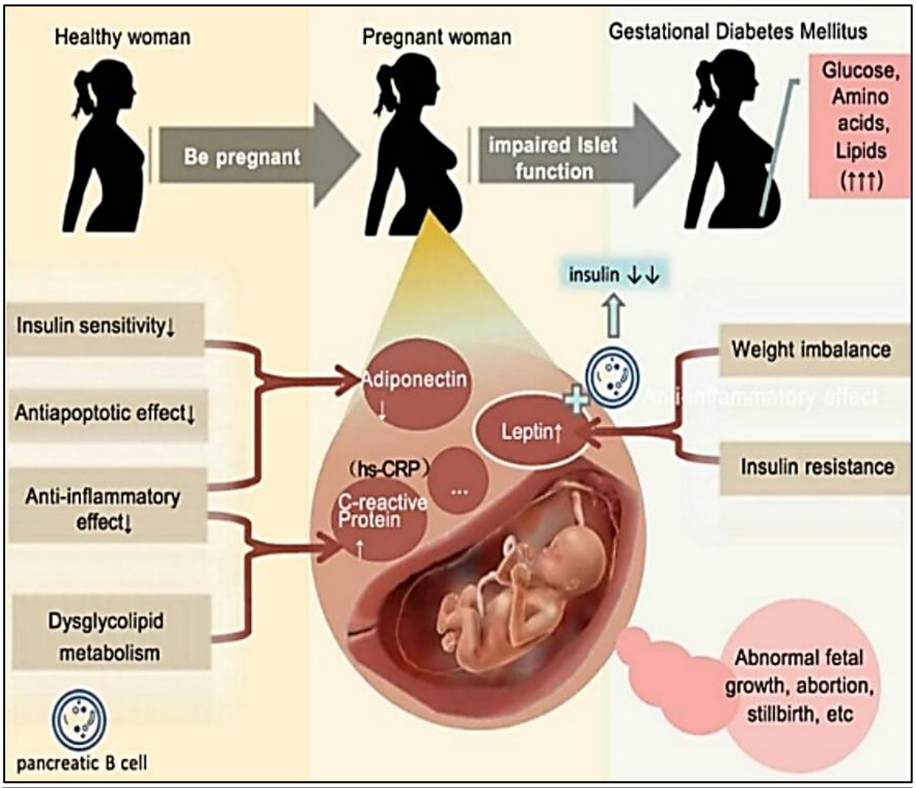
In this context, machine learning provides a powerful pathway to improve risk prediction and guide timely intervention strategies, ultimately contributing to better maternal health outcomes. The discussion here centres on predicting maternal health risks using previously recorded datasets, enabling the model to learn patterns and accurately estimate the maternal health status. The dataset includes crucial attributes such as age, blood pressure, blood sugar levels, body temperature, and heart rate key indicators strongly associated with maternal complications and mortality.

In this project, the primary focus is on identifying and addressing two major pregnancy-related conditions: preeclampsia and gestational diabetes. Both conditions pose serious risks to maternal and fetal health, and early detection is critical for effective management (Referred from the paper [4]). By analysing vital health parameters and learning from historical data, the machine learning model aims to detect the early signs of these conditions, supporting timely clinical decision-making and improving overall maternal care.



This diagram shows how placental dysfunction releases extracellular vesicles into the bloodstream, triggering endothelial damage, systemic inflammation, and coagulation abnormalities. These combined effects involve immune activation, impaired blood vessel function, and clotting disturbances, ultimately contributing to the development and progression of preeclampsia in pregnant women.

Figure 1.1 Preeclampsia



This diagram shows how pregnancy-related insulin decline and impaired islet function lead to gestational diabetes, causing insulin resistance, inflammation, weight imbalance, and increased fetal

Figure 1.2 Gestational diabetes

This project uses machine learning, specifically the XGBoost algorithm, to predict maternal health risks with high accuracy and efficiency. XGBoost identifies key patterns in vital parameters such as blood pressure, blood sugar, and heart rate, enabling early detection of complications. MySQL serves as the secure storage system, managing maternal records, physiological data, and prediction outputs with integrity and reliability. By integrating machine learning and structured data management, the system supports timely evaluation of maternal status. Such predictive tools can improve clinical decision-making, enhance early intervention, and contribute to reducing preventable maternal and neonatal complications.

Finally highlighting the importance of using machine learning to strengthen maternal healthcare, especially in high-risk regions. By integrating XGBoost predictions with secure MySQL data management, the system enables early detection of complications like preeclampsia and gestational diabetes, supporting timely interventions and helping reduce preventable maternal and neonatal deaths.

1.1 Problem Statement

"Visualizing maternal health risk analysis using the physiological data ($\boldsymbol{f_m}$) with XGBoost classification algorithm"

The formula is an model for estimating the level of maternal risk based on six physiological variables: Age, Systolic Blood Pressure (SystolicBP), Diastolic Blood Pressure (DiastolicBP), Blood Sugar level, Body Temperature, and Heart Rate (From the dataset). And the Risk Level as the target instance of the model. In this formula, $\boldsymbol{\hat{y}_m}$ represents the estimated maternal risk class, which is one of three: low, medium, or high risk.

The mapping function f_m is that of a trained XGBoost which is an ensemble machine learning algorithm that trains several decision trees in training and outputs the class mode of the classes output by the individual trees. The method utilizes the consensus decision-making of numerous trees to enhance predictive performance and mitigate overfitting.

$$\boldsymbol{\hat{y}_m = f_m(Age, SystolicBP, DiastolicBP, BS, BodyTemp, HeartRate)} \ \ldots\ (1)$$

In reality, the XGBoost model is trained on past patient data with these six features and their corresponding risk labels. Upon input of new patient data, the model analyses the values of these physiological parameters and estimates the risk level by combining the outputs of its individual trees. This allows clinicians to detect pregnancies with high-risk profiles early and apply preventive or monitoring actions accordingly.

This approach has been proven in various studies, demonstrating that XGBoost classifiers are highly accurate (typically >85%) for multiclass maternal risk classification problems using these vital physiological markers. The model's interpretability and robustness qualify it for clinical decision support systems to decrease maternal morbidity and mortality via timely risk stratification.

Therefore, the main problem addressed in this research is the development of an efficient, scalable machine learning model for predicting maternal health risk levels based on analysing key clinical variables collected during pregnancy. The objective is to enhance early risk identification by employing predictive analytics to enable better informed decision making for healthcare professional, thereby improving maternal morbidity and mortality rates. This problem involves problems such as handling unbalanced datasets, feature selection, optimization of model performance, and making the model applicable in actual clinical settings.

The model for estimating maternal risk levels based on six physiological variables typically uses multinomial logistic regression or ensemble machine learning frameworks. Key Formula Components

For multinomial logistic regression**:**

$$\ln\left(\frac{P(Low\ risk)}{P(High\ risk)}\right) = \beta_0 + \beta_1 Age + \beta_2 SystolicBP + \beta_3 DiastolicBP + \beta_4 BS + \beta_5 BodyTemp \quad \ldots (2)$$

$$\ln\left(\frac{P(Mid\ risk)}{P(High\ risk)}\right) = \gamma_0 + \gamma_1 Age + \cdots + \gamma_6 BodyTemp \quad \ldots (3)$$

The class with the highest probability $\hat{y}_m$ is selected as the risk level.

1.2 OBJECTIVES

1. To build a responsive maternal health website that ensures mobile access, simple navigation, visual aids, multilingual support, fast loading and accessible UI for rural populations.
2. Usage of supervised machine learning algorithm helps the model to learn from fresh data and identify intricate patterns.
3. To execute a model that customizes risk assessments to fit each individual’s maternal health profile.
4. To facilitate real-time input of data and provide timely health analysis allowing for quicker interventions and minimizing complications due to delayed treatment.

1.3 SCOPE OF THE STUDY

1. Development of a Maternal Risk Prediction Model using XGBoost – used because it delivers high accuracy, handles non-linear medical data effectively, and performs well even with smaller healthcare datasets.
2. Use of Python Data Science Stack i.e., Pandas, NumPy, Scikit-learn – adopted for efficient data cleaning, preprocessing, feature engineering, and model training due to its reliability and strong community support for healthcare analytics.

3. Visualization of Model Results using Matplotlib – employed to present feature importance, SHAP plots, and performance graphs in a clear and interpretable manner for clinical and academic review.

4. Data Storage and Management through MySQL – selected for secure structured storage of patient records and predictions, ensuring data consistency and scalability for healthcare systems.
5. Development of a Web-based Application using Flask – implemented to provide real-time access for healthcare workers, allowing easy input of patient vitals and instant risk assessment through a user-friendly interface.
6. Model Evaluation and Optimization Techniques – includes accuracy, F1-score, confusion matrix, and hyperparameter tuning to ensure a dependable model suitable for real clinical use.
7. Future-Ready Expansion Scope – scalable to integrate mobile apps, IoT health devices, and remote monitoring systems, enabling widespread use in hospitals, clinics, and rural maternal care programs.

1.4 ORGANIZATION OF THE REPORT

This report is structured systematically to provide a comprehensive understanding of the proposed project, ensuring clarity in each stage of its development. The organization is as follows:

Chapter 1: Introduction
Introduces the project by providing background context, clearly defining the main problem addressed, presenting the objectives pursued, and specifying the scope. It lays the foundation for understanding why maternal health risk prediction is crucial and outlines the aims and boundaries of the work undertaken.

Chapter 2: Literature survey
Reviews existing research, related technologies, and methodologies. It highlights previous efforts in maternal health risk prediction, comparing approaches, models, and systems. This assessment identifies gaps and informs the rationale for pursuing new solutions, ensuring the project builds upon established knowledge and addresses unresolved issues.

Chapter 3: Software requirements specification

Details both functional requirements (what the system should do) and non-functional criteria (performance, security). Defines user needs by analyzing target stakeholders' expectations. This chapter turns project ideas into concrete requirements, ensuring clarity and providing a baseline for the technical development of the system.

Chapter 4: Proposed methodology

Analysis current systems to pinpoint their shortcomings, then designs new components to address these gaps. It describes each system component, mapping requirements to practical solutions. The methodology transforms user needs and research insights into a systematic, efficient blueprint for project development.

Chapter 5: System architecture

Presents detailed diagrams and workflows, including the Data Flow Diagram that maps how information moves through the system. Defines the architecture, outlining modules, interfaces, and interactions. This structural overview supports technical implementation and clarifies how the Maternal Health Risk Prediction System will function.

Chapter 6: Application testing

Describes the process and results of unit testing performed on the application. Each component is tested independently to validate functionality, stability, and correctness.

Chapter 7: Interpretation of results

Showcases project deployment with images and descriptions that explain how the application operates.

Conclusion

Summarizes the project's accomplishments and the value delivered. Reflects on obstacles encountered and lessons learned throughout development. It discusses potential improvements and suggests future directions to enhance robustness, scalability, and impact, ensuring continued relevance and progress.

References

Lists all academic literature, research papers, and tools referenced in the project. Employs a standard citation style (IEEE or APA), ensuring transparent acknowledgment of prior work and supporting academic credibility.

Chapter 2

LITERATURE SURVEY

[1] L. Adeola, C. Iwendi and K. Akeren, "**Using Machine Learning to Predict and Monitor Maternal Health Risk: A Case Study of Materna – an End User Web Application**" Published on 2025(IEEE) AI-Driven Smart Healthcare for Society 5.0, Kolkata, India, 2025.

This research work sought to provide an ergonomic and easily accessible solution by creating a machine learning web application that will be able to detect maternal health with an accuracy of 84%. The web application is a simple form where the user has a set list of questions and answers, based on those choices, a prediction is made.

[2] N.R.G, S. V, K. G. R, V. T, A. S. Veluswamy and V. N, "**Unveiling the Deep Learning Techniques for Maternal Health Risk Prediction**," 2025 International Conference on Next Generation Computing Systems (ICNGCS), Coimbatore, India, 2025

Deep learning models, especially SAINT, outperform classical classifiers in predicting maternal health risks, offering improved accuracy, robustness, and potential for real-time, early intervention support.

[3] H. A. Shifa, M. U. Mojumdar, M. M. Rahman, N. R. Chakraborty and V. Gupta, "**Machine Learning Models for Maternal Health Risk Prediction based on Clinical Data**" Published on 2024(IEEE) 11th International Conference on Computing for Sustainable Global Development (INDIACom), New Delhi, India, 2024.

The study by Shifa et al. (2024) demonstrated that among various machine learning models applied to maternal health risk prediction, XGBoost and Random Forest achieved the highest accuracy, indicating their strong capability for effective risk assessment based on clinical data in real-world healthcare settings.

[4] "Machine learning-based maternal health risk prediction model for IoMT framework" Subhash Mondal, Amitava Nag, Anup Kumar Barman and Mithun Karmakar

Department of Computer Science and Engineering, Central Institute of Technology Kokrajhar, Kokrajhar, Assam, India, 2023.

IoT devices provide a means to collect real-time health data, enabling continuous monitoring and analysis in the Internet of Medical Things (IoMT) environments. By integrating IoT devices into the system, crucial signs such as Heart Rate (HR), Systolic and Diastolic Blood Pressure (BP), Fetal Movements (FM), and Temperature (T) can be tracked remotely and non-invasively. This allows for the timely detection of abnormalities or potential risk factors during pregnancy, empowering healthcare professionals to intervene proactively and provide personalized care.

[5] Yang Y, Wu N. **“Gestational Diabetes Mellitus and Preeclampsia: Correlation and Influencing Factors.”** Front Cardiovasc Med. 2022 Feb 16;9:831297. doi: 10.3389/fcvm.2022.831297. PMID: 35252402; PMCID: PMC8889031.

Research indicates a strong association between gestational diabetes mellitus (GDM) and preeclampsia (PE), as both conditions share common risk factors and physiological changes. Several studies show that women with GDM have a higher likelihood of developing PE, and the combination of both disorders increases the risk of adverse perinatal outcomes. Evidence suggests that GDM is independently linked to PE, and controlling GDM through effective management can significantly lower the incidence of PE. Identifying controllable factors, understanding the underlying mechanisms, and recognizing predictive markers are essential for improving maternal and fetal health outcomes.

[6] M. Ahmed and M. A. Kashem, **"IoT Based Risk Level Prediction Model For Maternal Health Care In The Context Of Bangladesh"** Published on 2020(IEEE) 2nd International Conference on Sustainable Technologies for Industry 4.0 (STI), Dhaka, Bangladesh, 2020.

This research, a system has been developed for effective monitoring and predicting risk level of a pregnant women, in the context of Bangladesh. For the analysis of risk factors, categorize and classifying approaches has been used according to the intensity of risk. After comparing among some groups of the machine learning algorithm, in case of classification and prediction of the risk level shows that Modified Decision Tree Algorithm gives the highest accuracy and the numeric value of this accuracy is 97%.

[7] S.S.L.Pereira et al., "Improving Maternal Risk Analysis in Public Health Systems" Published on *2020*(IEEE) 5th International Conference on Smart and Sustainable Technologies (SpliTech), Split, Croatia, 2020.

This study applies the Recursive Feature Elimination (RFE) strategy associated with decision tree-based classifier identifying a relevant set of features among an extensive list of maternal predictive information considered in a decision-making process.

[8] "Machine learning applied in maternal and fetal health: a narrative review focused on pregnancy diseases and complications" By Lawrence Merle Nelson Mary Elizabeth Conover Foundation, Inc., United States.

The applications are varied and point not only to the diagnosis, but also to the management, treatment, and pathophysiological understanding of perinatal alterations. Facing the challenges that come with working with different types of data, the handling of increasingly large amounts of information, the development of emerging technologies, and the need of translational studies, it is expected that the use of ML continue growing in the field of obstetrics and gynaecology.

[9] A. Maitra and N. Kuntagod, "**A novel mobile application to assist maternal health workers in rural India**," 2013 5th International Workshop on Software Engineering in Health Care (SEHC), San Francisco, CA, USA, 2013.

This mobile application for maternal health enables rural caregivers to manage pregnancy information and receive automated risk assessments and care advice. Operating on edge devices, it minimizes network dependence and medical expertise, supporting treatment adherence and early intervention, even before professional medical help is available. Deployment shows improved care outcomes.

Chapter 3

SOFTWARE REQUIREMENTS SPECIFICATION

3.1 FUNCTIONAL REQUIREMENTS

1. User Input Interface

The system shall provide an intuitive and user-friendly interface that allows users (patients or healthcare professionals) to input essential clinical parameters required for risk analysis. The input fields shall include:

- Age (years): Represents the age of the pregnant individual.
- Diastolic Blood Pressure (mmHg): Indicates the lower pressure in arteries when the heart rests between beats.
- Blood Sugar (mmol/L): Reflects the glucose concentration in the blood, helping in identifying diabetes or gestational diabetes.
- Body Temperature (°C): Detects fever or abnormal temperature levels which may indicate infection or metabolic issues.
- Heart Rate (bpm): Measures beats per minute to assess cardiovascular health during pregnancy.

2. Risk prediction

The system shall use a pre-trained supervised machine learning model with XGBoost to process the input parameters and predict the maternal health risk level. The model shall analyse the clinical inputs and generate a risk category output based on learned patterns from the dataset.

3. Risk classification output

The system shall classify the maternal health condition into the following risk categories:

- Low Risk: Normal and healthy pregnancy status.
- Medium Risk: Requires periodic monitoring and clinical observation.
- High Risk: Indicates potential complications and requires immediate medical attention.

4. Result display and visualization

The system shall visually display the prediction result using color-coded indicators for better interpretation:

- 🟢 Green: Low Risk
- 🟠 Orange: Medium Risk
- 🔴 Red: High Risk

Additionally, the system shall provide interactive visualizations like line charts and bar charts to compare the patient's health parameters over time. These visualizations will help clinicians track physiological trends across pregnancy months and identify risk progression or improvement.

5. Result storage and data management

The system shall store each patient's input data, prediction results, recommendation and visualization summaries in a secure database like MySQL.
Stored data can be retrieved for:

- Tracking patient progress across trimesters.
- Generating medical reports.
- Supporting clinical decision-making and longitudinal studies.

Data storage shall ensure privacy, integrity, and accessibility for authorized users only

3.2 NON-FUNCTIONAL REQUIREMENTS

The Maternal Health Risk Prediction System must ensure strong security, high performance, scalability, and user-friendly operation to support accurate clinical decision-making. It should run efficiently in real time, remain reliable under growing data volumes, and be easy to maintain for future updates. The design must prioritize fast response, protected data handling, and seamless integration with hospital systems. Interoperability with healthcare databases and adaptability to evolving medical standards are essential for long-term accuracy and usability.

3.2.1 Performance requirements

- **Prediction speed:**
  The system must generate maternal health risk predictions within 1 second after submission, ensuring a smooth and responsive user experience for healthcare professionals.
- **Accuracy:**
  The machine learning model must achieve a minimum prediction accuracy of 87%, verified through testing with certified and validated clinical datasets.
- **Algorithm efficiency:**
  The prediction model shall be developed using a high-performance ensemble algorithm such as XGBoost, Extra Trees Classifier, or Random Forest, ensuring robustness, interpretability, and reliability in medical data processing.
- **System responsiveness:**
  The user interface should maintain optimal responsiveness and handle multiple concurrent user sessions without any degradation in speed, prediction accuracy, or visualization quality.

3.2.2 Supportability requirements

- **Technology stack:**
  - Python 3.13.3: Primary development language for implementing machine learning and backend logic.
  - Flask: Used to build a lightweight, interactive web interface for user input and result display.
  - Pickle: Utilized to load and deploy the trained ML model from a .sav or serialized file.
  - Pandas: Handles structured clinical data input, preprocessing, and output in tabular format.
  - Matplotlib: Provides visual analytics and data plotting functionalities for patient monitoring.

- **Modularity:**
  The software architecture shall be modular, enabling easy updates, replacement, or retraining of the ML model without affecting other components of the system such as the interface or database.

- **Platform compatibility:**
  The application shall be fully compatible with Windows, Linux, and macOS operating systems.

- **Maintainability and scalability:**
  The system must support continuous updates for new datasets and medical standards. Its design should allow for scalable storage expansion as patient data grows, ensuring long-term sustainability and performance stability.

Software Requirements:	**Hardware Requirements:**
Python.	Processor: Minimum Dual-core CPU.
Flask.	RAM: 4 GB or higher.
HTML, CSS and JavaScript.	Storage: At least 1 GB free space.
Pickle.	
Pandas.	
Sci-kit learn.	
IDE (VS Code and Jupiter notebook).	
Browser (Chrome).	
Operating System (Windows).	

Table 3.1 Software and hardware requirements

Chapter 4

PROPOSED METHODOLOGY

4.1 MOTIVATION

Maternal health remains a major public health concern, especially in developing and rural regions where delayed assessments, inconsistent monitoring, and limited access to healthcare facilities contribute to increased maternal morbidity and mortality. Conditions such as preeclampsia (characterized by dangerously high blood pressure during pregnancy) and gestational diabetes (abnormal glucose levels that develop during pregnancy) are among the leading causes of complications for both mothers and infants. These conditions often progress silently and are detected only after significant harm has occurred, primarily due to irregular checkups and delayed diagnosis.

This project aims to address these critical challenges through the integration of supervised machine learning for continuous, personalized, and data-driven maternal risk evaluation. By leveraging machine learning algorithms, the system can identify early and subtle patterns in clinical parameters—such as blood pressure, blood sugar, heart rate, and body temperature that traditional manual assessments might overlook. Early detection of anomalies enables healthcare professionals to intervene promptly, minimizing complications arising from preeclampsia, gestational diabetes, or other pregnancy-related risks.

In addition, the system's scalability ensures it can handle growing datasets, multiple users and future integration with digital health platforms, electronic medical records, and IoT-enabled monitoring devices. Beyond individual benefits this approach contributes to preventive healthcare, supporting early intervention, reducing maternal emergencies and promoting equitable access to quality maternal care.

By combining the power of artificial intelligence with continuous clinical monitoring, this project not only strengthens maternal healthcare systems but also aligns with the broader vision of digital health innovation. It supports clinicians with data-driven insights, reduces manual workload, and lays the foundation for future advancements in maternal wellness, ultimately working toward safer pregnancies and healthier outcomes for mothers and children alike.

4.2 EXISTING SYSTEM

Existing maternal health risk prediction systems primarily rely on supervised machine learning techniques to classify pregnancy risk levels and support early clinical intervention. These systems use datasets collected from hospitals, health centres and community surveys consisting of key clinical parameters such as maternal age, blood pressure, heart rate, blood glucose levels, body temperature, and medical history. Models such as Logistic Regression, Random Forest, Support Vector Machines, K-Nearest Neighbours, Naive Bayes, Neural Networks, and particularly XGBoost have been widely used due to their ability to learn predictive patterns from structured clinical data. Several studies have compared multiple algorithms to identify the most effective model, with XGBoost achieving high accuracy—up to 97.3% in some cases—demonstrating strong potential for accurate maternal risk classification.

Some existing solutions go beyond research and offer basic application-based tools. For example, lightweight web applications allow users to input clinical data and receive a predicted risk level. While such tools provide accessible maternal risk assessment, they function mostly as direct input-output systems without the ability to store patient records, monitor clinical trends, or support large-scale deployment across healthcare networks. This limits their long-term use in real clinical environments.

4.2.1 Challenges in Existing Methodology

1. **Absence of patient data storage and follow-up tracking**
   Existing models provide only one-time predictions with no provision to store patient information. The proposed project integrates a secure database to maintain patient records, enabling continuous monitoring of maternal health across multiple visits and supporting long-term clinical assessment.
2. **Lack of visual dashboards for clinical decision support**
   Current systems do not provide visual insights on patient progress or risk patterns. The proposed solution includes interactive data visualization tools such as trend graphs, risk comparison charts, and analytics dashboards to help healthcare professionals quickly interpret results and take timely actions.

3. **Limited real-world deployment and scalability**
   Most existing approaches are standalone and not suitable for large-scale use. The proposed system is designed with a scalable architecture that can be deployed across hospitals, clinics, and rural health centres, ensuring broader accessibility and impact.
4. **No support for comparative or population-level analytics**
   Existing tools fail to analyse patterns across communities. The new system enables aggregate analytics to identify common risk factors, regional trends, and population-level maternal health risk distributions, aiding government and public health planning.
5. **Lack of explainability in model predictions**
   Most models provide predictions without explaining the reasoning behind them, reducing trust among clinicians. The proposed solution integrates Explainable AI (XAI) through SHAP, allowing doctors to understand feature contributions behind each prediction and improving acceptance in clinical use.
6. **Minimal integration into clinical workflows**
   Existing systems do not align with healthcare professionals' operational needs. The proposed project includes a user-friendly interface, role-based access, and record management features that support real clinical workflows and encourage adoption.

4.3 PROPOSED SYSTEM

The proposed system aims to develop an advanced, scalable, and intelligent Maternal Health Risk Prediction and Monitoring platform that overcomes the limitations found in existing methodologies. This system integrates machine learning, Explainable AI, data visualization, and database-driven patient management to provide accurate risk classification and continuous monitoring for pregnant women. The solution is designed not only for predictive analysis but also to support clinical decision-making, long-term tracking, and population-level health insights. At the core of the proposed system is a machine learning model that classifies maternal risk into low, medium, or high categories. The model utilizes six key physiological parameters — Age, Systolic Blood Pressure, Diastolic Blood Pressure, Blood Sugar Level, Body Temperature, and Heart Rate as input features. Advanced algorithms such as XGBoost are employed due to their high accuracy, robustness, and efficiency in handling clinical datasets (from the reference paper [2]).

XGBoost enhances prediction capability by combining multiple decision trees through a boosted ensemble approach, reducing overfitting and improving generalization.

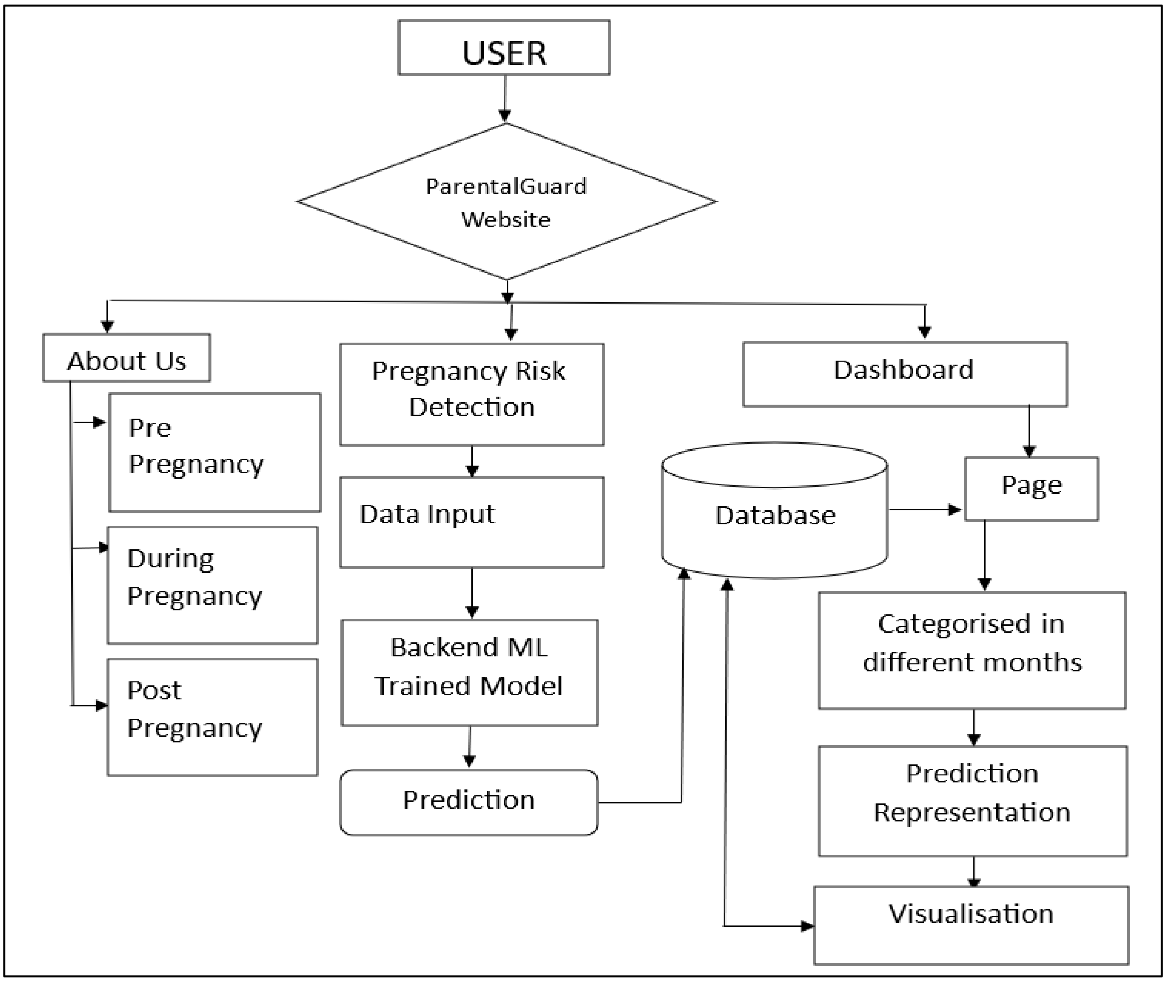


Figure 4.1 Flow chart of the proposed system

A major advancement of the proposed system is the integration of Explainable AI (XAI) using SHAP (SHapley Additive exPlanations). While many existing systems predict risk without justification SHAP provides clear insights into how each feature influences the model's output. This transparency enables healthcare practitioners to trust system recommendations, understand risk-driving factors, and make informed clinical decisions. The inclusion of XAI also enhances the ethical and safe use of AI in healthcare.

Unlike current one-time prediction models, the proposed system incorporates a centralized database using MySQL. This allows secure storage of patient records, repeated assessment data, and historical health trends. Healthcare workers can track patient progress across multiple visits, enabling proactive intervention and personalized care. Data encryption and role-based access control ensure privacy and compliance with medical data protection standards.

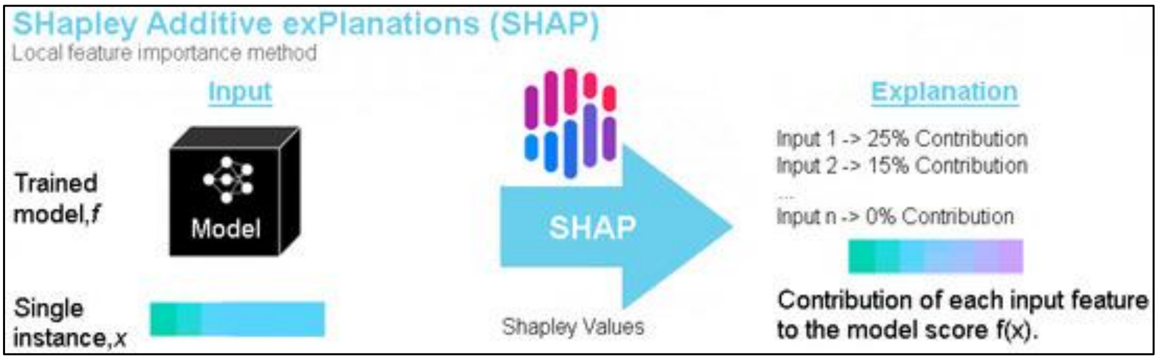


Figure 4.2 SHAP Explanation AI for the risk, additional explanation and evaluation.

To enhance usability, the system includes a user-friendly web interface developed using Flask or Django for backend integration, and HTML, CSS, and JavaScript for frontend interactions (from the reference paper [1]). The interface allows doctors, nurses, and health workers to input patient details, view predicted risk levels, analyze previous reports. For remote or rural healthcare applications, a lightweight responsive design ensures compatibility with tablets and smartphones.

4.4 SUPERVISED MACHINE LEARNING

Supervised Machine Learning is a type of machine learning approach where models are trained using labelled datasets. In this method, each input data point is associated with a corresponding output label, allowing the model to learn the mapping between inputs and outputs. The algorithm analyses patterns within the data to predict outcomes for new, unseen inputs. Supervised learning is widely used for tasks such as classification, regression, and risk prediction. Common examples include predicting disease outcomes, credit scoring, and spam detection. The main goal is to minimize the error between predicted and actual results through continuous learning and optimization. XGBoost (Extreme Gradient Boosting) is an advanced supervised machine learning algorithm based on the gradient boosting framework (improvised from the reference papers). It builds an ensemble of weak decision trees sequentially, where each tree corrects the errors of the previous one. XGBoost is known for its high accuracy, efficiency, and scalability. It supports both classification and regression tasks and handles missing data effectively. The algorithm optimizes performance through regularization techniques, preventing overfitting and improving generalization. In healthcare applications, such as maternal health risk prediction, XGBoost effectively identifies complex, non-linear relationships among clinical parameters, providing accurate risk classification and valuable insights for timely medical decision-making.

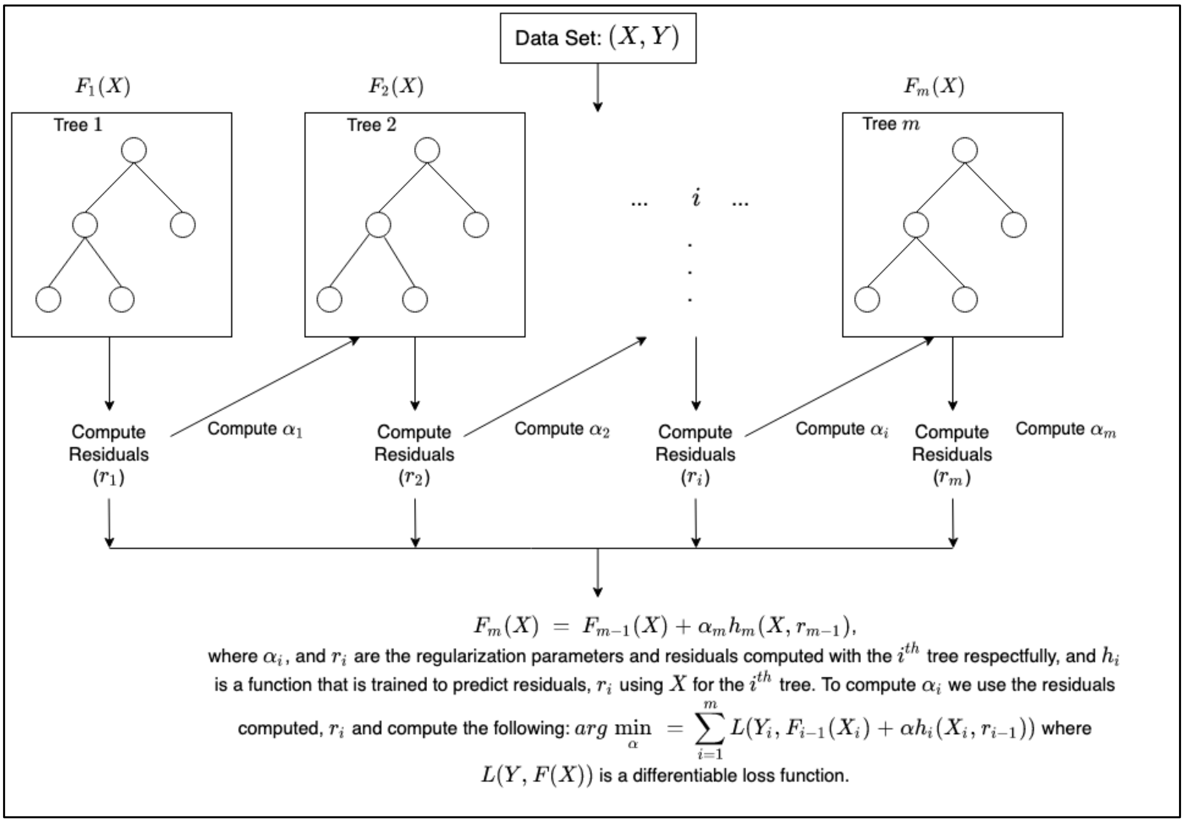


Figure 4.3 Understanding the XGBoost from the roots.

The proposed system also addresses the lack of visualization in current solutions by offering interactive dashboards. These dashboards display patient-specific trends, population-level analytics, feature importance graphs, SHAP visualizations, and comparisons of risk patterns across time or regions. Such visual tools support faster decision-making, improve clinical efficiency, and help identify emerging maternal health risk trends at community or district levels.

Overall, the proposed system provides a comprehensive, explainable, and scalable solution for maternal risk prediction. It enhances early diagnosis, supports continuous maternal care, improves clinical decision-making, and contributes to reducing maternal morbidity and mortality through intelligent technology-driven healthcare.

4.5 RESULT PREDICTION

A commonly used prediction algorithm for maternal risk classification is multinomial logistic regression, which estimates the probability of each risk class (low, medium, high) based on physiological variables. The general equation for predicting the probability that a case belongs to class k (where k is one of the risk levels) is:

$$\frac{P(y=k) = \exp(\beta_{k0} + \beta_{k1}\text{Age} + \beta_{k2}\text{SystolicBP} + \beta_{k3}\text{DiastolicBP} + \beta_{k4}\text{BS} + \beta_{k5}\text{BodyTemp} + \beta_{k6}\text{HeartRate})}{\sum_{j=1}^{3}\exp(\beta_{j0} + \beta_{j1}\text{Age} + \cdots + \beta_{j6}\text{HeartRate})} \quad \ldots (4)$$

Where:

- $\boldsymbol{\beta_{ki}}$ are the coefficients for each variable and risk class,
- $\boldsymbol{y}$ is the predicted risk class (low, medium, high),
- The denominator sums over all three risk classes to ensure probabilities add up to.

The predicted class $\hat{\boldsymbol{y}}_{\boldsymbol{m}}$ (from the equation (1)) is the one with the highest probability.

4.6 PRE-PROCESSING

Pre-processing is a crucial phase that ensures the dataset used for maternal health risk prediction is accurate, consistent, and suitable for machine learning modelling. It involves a systematic pipeline of data collection, cleaning, transformation, model training, and backend integration to deliver a reliable prediction system. We have performed the following pre-processing of the data to ensure that dataset is reliable:

1. Data Collection

The first stage involves gathering maternal health-related data from trusted medical and public health sources. The dataset typically consists of clinical parameters such as age, blood pressure, blood glucose levels, body temperature, heart rate, and other maternal health indicators. Data can be sourced from hospitals, research repositories, government health portals, or open-source datasets. The aim is to obtain a diverse and representative dataset that reflects different risk profiles of pregnant women. Once collected, the dataset is stored in a structured format such as CSV or Excel for initial processing. Proper documentation of attributes, units of measurement, and data acquisition methods is maintained to ensure standardization.

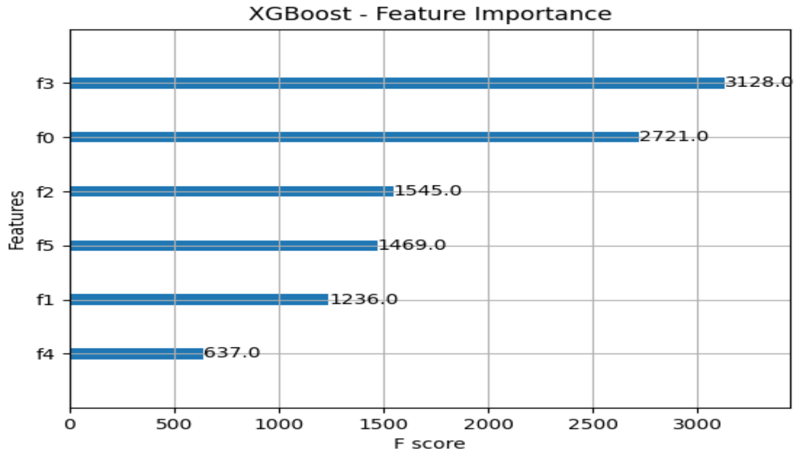


Figure 4.4 Importance of each feature in the dataset.

This figure 4.4 chart shows which features the XGBoost model relied on the most while making predictions.

- f3(Blood Glucose Level) and f0(Systolic or Diastolic Blood Pressure) have the highest scores, meaning they contributed most to the model's decision-making.
- Features with higher F-scores were used more frequently across decision trees, indicating stronger influence.
- This helps us understand which input factors are most critical for risk classification and supports feature selection or clinical interpretation.
- Features with lower F-scores were still used by the model but had comparatively weaker correlations with risk outcomes.

2. Data Cleaning and Preparation

Data cleaning improves the quality and usability of the dataset. This step begins with identifying missing values, duplicate entries, and inconsistent records. Missing values are handled through either removal or imputation techniques, depending on their frequency and impact. Outlier detection is performed using statistical methods to identify abnormal or clinically impossible readings. Data normalization or scaling is applied to ensure uniform value distribution, especially for features used in the XGBoost model. Categorical attributes, if any, are encoded into numerical formats. Correlation analysis is also carried out to understand the influence of each feature on maternal health outcomes. The cleaned and structured dataset is then split into training and testing subsets for machine learning.

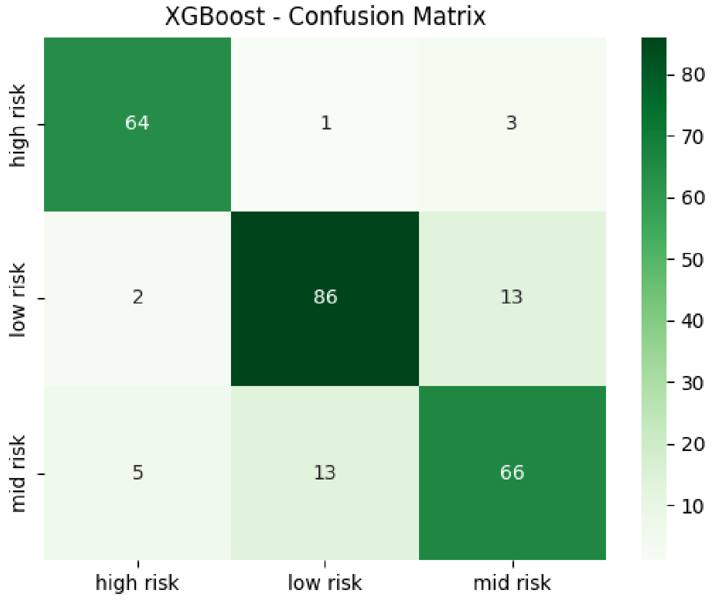


Figure 4.5 Confusion matrix of the XGBoost trained model.

3. Machine Learning Model Training with XGBoost

Once the dataset is pre-processed, the machine learning pipeline is initiated. XGBoost (Extreme Gradient Boosting) is selected due to its accuracy, robustness, and capability to handle complex clinical data. The training process includes feature selection, model hyperparameter tuning, and iterative training to enhance accuracy and minimize prediction error. Evaluation metrics such as accuracy, precision, recall, F1 score, and confusion matrix are used to assess performance. Model interpretability is incorporated using techniques like SHAP values to understand feature importance, making the system more transparent and suitable for healthcare applications. The finalized model is then serialized and saved for backend integration.

4. Backend and Database Setup

For deployment, a backend system is developed using a suitable framework such as Flask or Django. The trained XGBoost model is integrated with the backend to enable real-time prediction based on user inputs. A relational database i.e., MySQL is configured to store patient data, prediction history, and model logs. The database schema includes structured tables for user details, clinical parameters, risk results, and recommendations. APIs are built to manage data flow between the frontend, backend, and the model. Security and privacy measures are implemented to ensure data confidentiality.

4.7 ABOUT DATASET

The Maternal Health Risk Dataset provides valuable clinical information about pregnant women, making it an excellent resource for predicting maternal health risks in a reliable and informed manner. It captures essential physiological parameters that reflect the overall health condition of expecting mothers. Each entry in the dataset represents an individual patient and includes measurable indicators such as age, blood pressure, blood sugar, body temperature, and heart rate. These factors are commonly analysed in healthcare to assess maternal well-being, as they can significantly influence the outcome of pregnancy and identify early warning signs of complications.

One of the key strengths of this dataset is its focus on medically relevant attributes directly linked to maternal health. Since the data is well-structured and clearly labelled into three categories—Low, Medium, and High risk—it becomes extremely useful for machine learning applications. Models like XGBoost can efficiently learn from these patterns and deliver accurate predictions regarding the level of maternal risk. This empowers healthcare professionals with data-driven insights to support timely diagnosis, preventive care, and better clinical decisions.

```
PS C:\Users\Mallikarjun K\OneDrive\Desktop\major_project\PrenatalGuard> python XGboost.py
   Age  SystolicBP  DiastolicBP    BS  BodyTemp  HeartRate  RiskLevel
0   25         130           80  15.0      98.0         86  high risk
1   35         140           90  13.0      98.0         70  high risk
2   29          90           70   8.0     100.0         80  high risk
3   30         140           85   7.0      98.0         70  high risk
4   35         120           60   6.1      98.0         76   low risk
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1014 entries, 0 to 1013
Data columns (total 7 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   Age          1014 non-null   int64
 1   SystolicBP   1014 non-null   int64
 2   DiastolicBP  1014 non-null   int64
 3   BS           1014 non-null   float64
 4   BodyTemp     1014 non-null   float64
 5   HeartRate    1014 non-null   int64
 6   RiskLevel    1014 non-null   object
dtypes: float64(2), int64(4), object(1)
memory usage: 55.6+ KB
None
```

Figure 4.6 The description of the dataset.

The dataset is also balanced and exhibits a healthy variation in clinical values, which enhances the model's learning capability. A balanced dataset ensures that the model performs equally well across all risk categories, avoiding bias toward one specific class. This makes the dataset highly suitable not only for academic research but also for practical, real-time maternal health monitoring systems.

Overall, the Maternal Health Risk Dataset holds immense potential for improving healthcare outcomes. By enabling predictive analytics, early risk detection, and continuous monitoring, it supports the development of intelligent maternal care systems that could ultimately contribute to safer pregnancies and healthier lives for mothers and babies. The dataset typically includes the following attributes:

Feature	Description
Age	Age of the woman during pregnancy, as risk increases with maternal age.
Systolic Blood Pressure	Upper value of blood pressure; higher SBP may indicate hypertension.
Diastolic Blood Pressure	Lower value of blood pressure; essential for assessing preeclampsia risk.
Blood Glucose Level	Indicates the possibility of gestational diabetes.
Body Temperature	Helps detect infection or abnormal metabolic conditions.
Heart Rate	Reflects cardiovascular health and potential complications.
Risk Level	Target variable classified as Low, Medium, or High risk based on health indicators.

Table 4.1 The description of dataset

4.8 MySQL STORAGE SYSTEM

In this system, patient data is efficiently managed using a MySQL database structured into three schemas: Trimester1, Trimester2, and Trimester3, reflecting the clinical stages of pregnancy. Each schema contains three monthly tables that store detailed health records for each month, such as Table1–Table3 for Trimester1 and Table4–Table6 for Trimester2. Trimester3 similarly includes Table7–Table9. Additionally, Table10 and Table11 store consolidated risk summaries, diagnostic outcomes, and combined longitudinal data used by the prediction model. This organized storage design ensures accurate month-wise tracking, supports trimester-level analysis, and enables secure, fast retrieval for effective maternal health risk prediction.

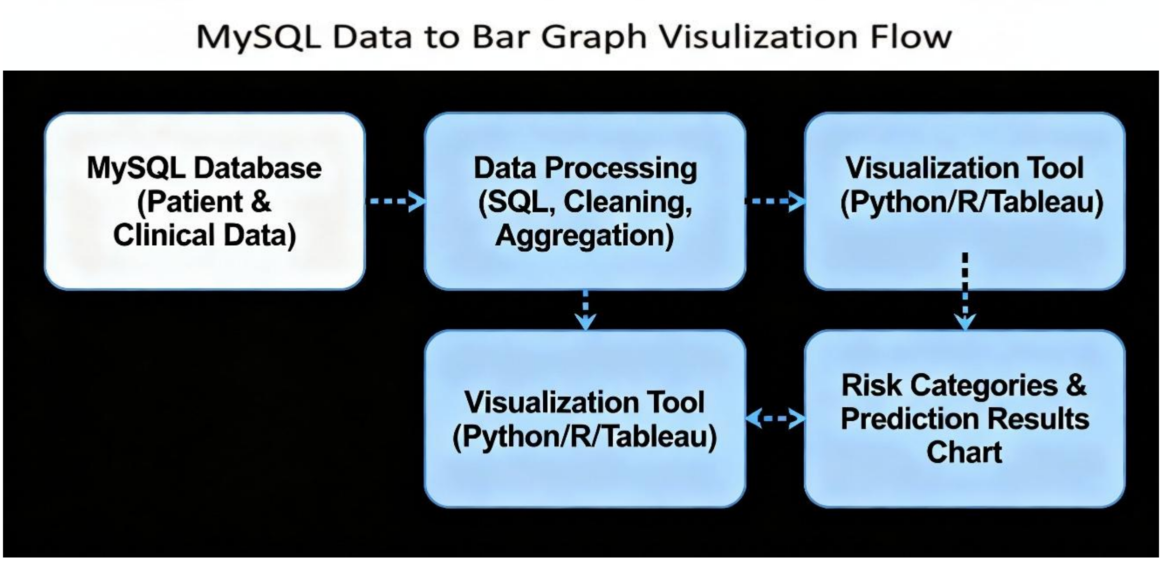


Figure 4.7 MySQL database usage in the project.

4.9 VISUALIZATION

The presented graphs shown in the implementations how key maternal health parameters vary across different months of pregnancy, offering a clear picture of physiological changes and their impact on risk levels. The monitored parameters include systolic and diastolic blood pressure, blood sugar, body temperature, and heart rate—each essential for assessing maternal well-being. In the early pregnancy stage, particularly Month 1 and Month 2, both blood pressure readings appear significantly elevated, indicating hypertension and resulting in a high-risk classification. By Month 3, notable improvements in blood pressure and heart rate reflect a shift to a low-risk condition as the maternal body gradually adjusts to pregnancy. Month 4 displays a moderate rise in temperature and blood pressure, placing the patient in a mid-risk zone due to increased metabolic and cardiovascular demands from fetal development. The bar chart highlights the magnitude of each parameter, while the line chart illustrates the overall trend. Together, these graphs support effective monitoring, early detection of complications, and timely clinical intervention.

Chapter 5

SYSTEM ARCHITECTURE

The PrenatalGuard system is a three-tier web architecture:

(1) Clients, via a responsive Frontend UI like web-mobile browsers, field devices, send in patient vitals (Age, Systolic/Diastolic BP, Blood Sugar, Temperature, Heart Rate).

(2) The requests route to a Flask REST API that enforces Auth & RBAC and runs a Preprocessing module (Pandas/NumPy) that will validate and normalize these inputs and subsequently invoke the Model Service, which loads the trained XGBoost model, serialized with pickle, to generate a Low/Medium/High risk prediction.

Layer	Components	Description
Clients	Web Browser, Field Worker Device, IoT Device	These are the entry points for users and data. They initiate requests to /api/predict to the backend.
Frontend	PrenatalGuard UI	Renders the interface for data input, dashboards, and visualization results.
Backend (Python Flask)	API, Auth, Preproc, ModelSvc, XAI, VizGen	This is the core application logic. The Flask REST API handles routing, delegates tasks to internal services (like Preprocessing and Model Service), and coordinates the workflow, including calling the XAI Service (SHAP) for explainable predictions.
Persistence	MySQL DB, Model Store, Logs	Stores structured data (MySQL for patient profiles and risk summaries), machine learning artifacts (Model Store for .sav model files and scalers), and tracks system activity (App/Audit Logs).
Admin & Ops	Admin Console, Monitoring	Provides tools for system management (Admin Console for user/model management) and maintenance (Monitoring for performance and error alerts).

Table 5.1 System architecture breakdown

(3) To add transparency to each of these predictions, they are routed to an XAI service (SHAP) that returns feature-level explanations, while the API will also invoke a Visualization generator (Matplotlib) to create charts and time-series views. All inputs are saved, along with prediction outputs and visual artifacts, into a MySQL database, organized exactly as in your doc, with schemas for Trimester1/Trimester2/Trimester3, each containing monthly tables and summary tables, and written to audit logs.

An Admin Console will allow model redeploys via ModelStore, user management, and system monitoring. This layout will support real-time predictions ≤1s target, longitudinal tracking, explainability, and future IoMT / EMR integrations.

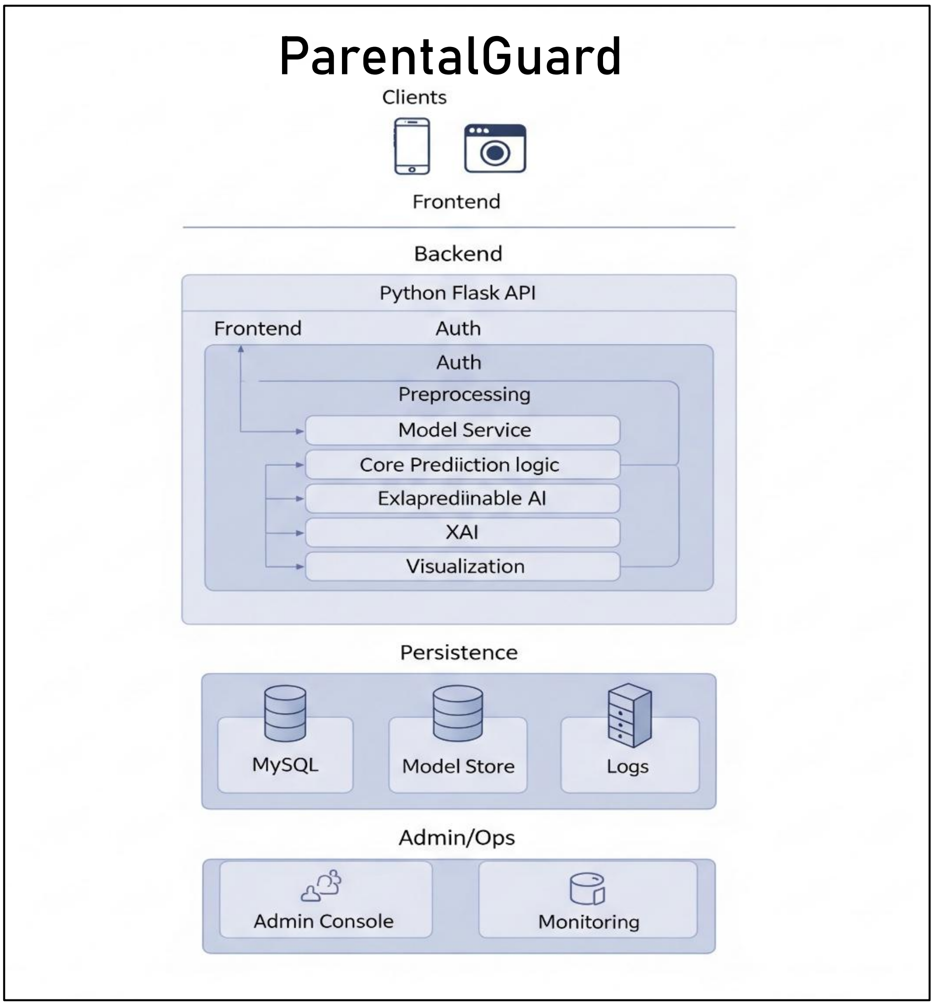


Figure 5.1 System architecture of parentalGuard

5.1 DATA FLOW DIAGRAM (DFD)

The Data Flow Diagram illustrates the flow of information through the Maternal Health Risk Prediction System. It provides a structured view of how input data is collected, processed, stored, and transformed into meaningful risk predictions for maternal health assessment. The DFD is represented across three levels to progressively detail the system's operations.

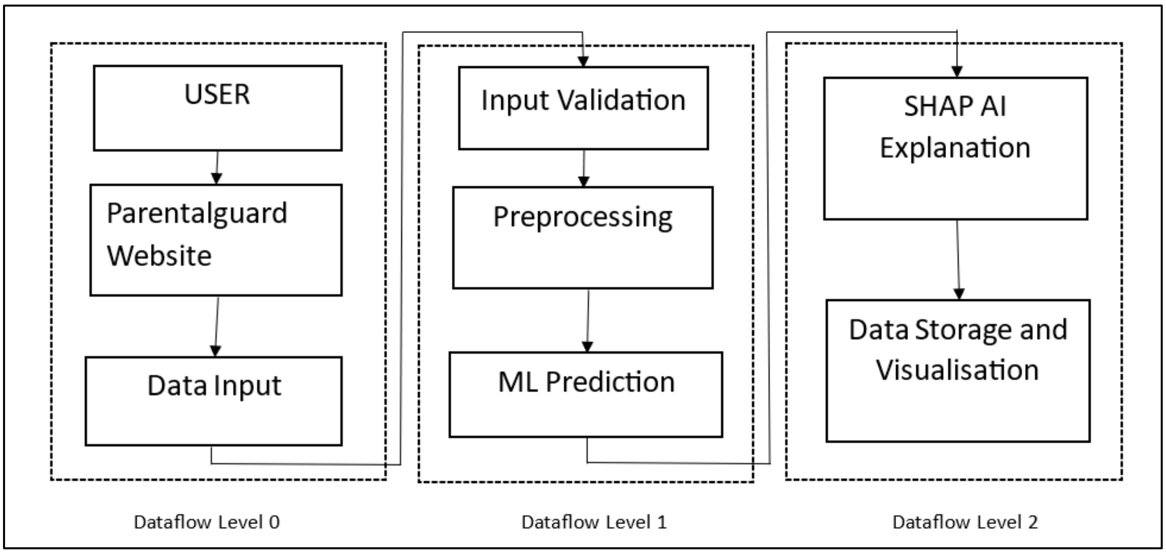


Figure 5.2 Dataflow diagram

5.1.1 DFD Level 0 – Context Diagram

The Level-0 DFD provides a high-level overview of the system and its interaction with external users. In this level, the entire system is represented as a single process known as the Maternal Health Risk Prediction System. The primary external entity interacting with the system is the User, who may be a healthcare worker, doctor, or patient. The user provides the maternal clinical inputs such as Age, Blood Pressure, Blood Sugar, Body Temperature, and Heart Rate. The system processes this information and produces a Maternal Risk Report that classifies the risk as Low, Medium, or High. This level emphasizes the core purpose of the system: accepting clinical data and returning a risk prediction without exposing the internal processing steps.

5.1.2 DFD Level 1 – System-Level Dataflow

The Level-1 DFD decomposes the single high-level process into key functional modules to show how data moves within the system. The major internal processes include:

1. Data Input and Validation – The system initially captures the maternal health parameters entered by the user and validates them to ensure completeness, format correctness, and value consistency.
2. Data Pre-processing – The validated data is cleaned and transformed to remove inconsistencies and prepare it for accurate model prediction. This may include normalization or handling missing values.
3. Database Interaction – Patient records are stored, retrieved, or updated in the database for future reference and historical tracking.
4. Machine Learning-Based Risk Prediction – The preprocessed data is fed into the trained XGBoost model, which analyzes the clinical patterns and predicts the maternal risk category.
5. Result Generation and Visualization – The prediction result is formatted into a clear output that may include textual results, color-coded risk levels, or visual charts to support medical interpretation.

This level clarifies how input and output data are internally transformed through sequential processes.

5.1.3 DFD Level 2 – Detailed Process Flow

The Level-2 DFD provides a more detailed breakdown of the key processes identified in Level-1, focusing on internal sub-processes and data refinement steps.

- Data Pre-processing Module is further divided into data cleaning, normalization, and feature selection to enhance model input quality. Data cleaning handles missing or incorrect values, normalization standardizes clinical units, and feature selection ensures that only the most relevant maternal health indicators are considered.
- Database Module includes sub-processes for storing new patient data, retrieving existing records for follow-up assessments, and updating clinical information for longitudinal patient monitoring. This enables continuity of maternal care and supports repeated evaluations over time.
- Machine Learning Prediction Module includes loading the trained XGBoost model, mapping the input clinical features to the model, generating the maternal risk prediction, and optionally producing interpretability insights such as feature importance

or SHAP values. This level reflects how the system transforms data into evidence-based decision outputs.

Level-2 ensures that every critical function is shown with detailed data movement, making the system transparent, traceable, and implementation-ready.

5.2 USE CASE DIAGRAM

The Use Case Diagram illustrates how different actors interact with the Maternal Health Risk Prediction System and the functions they access. The User enters personal and clinical data required for risk evaluation, forming the foundation for prediction. The Doctor reviews risk reports, monitors patient history, and uses the system's insights for clinical decision-making. The System Administrator oversees user accounts, updates configurations, and ensures smooth operation of the model and database. The Machine Learning Model acts as an internal system component that processes patient inputs to generate risk predictions. Together, these interactions support efficient maternal risk assessment and streamlined healthcare management.

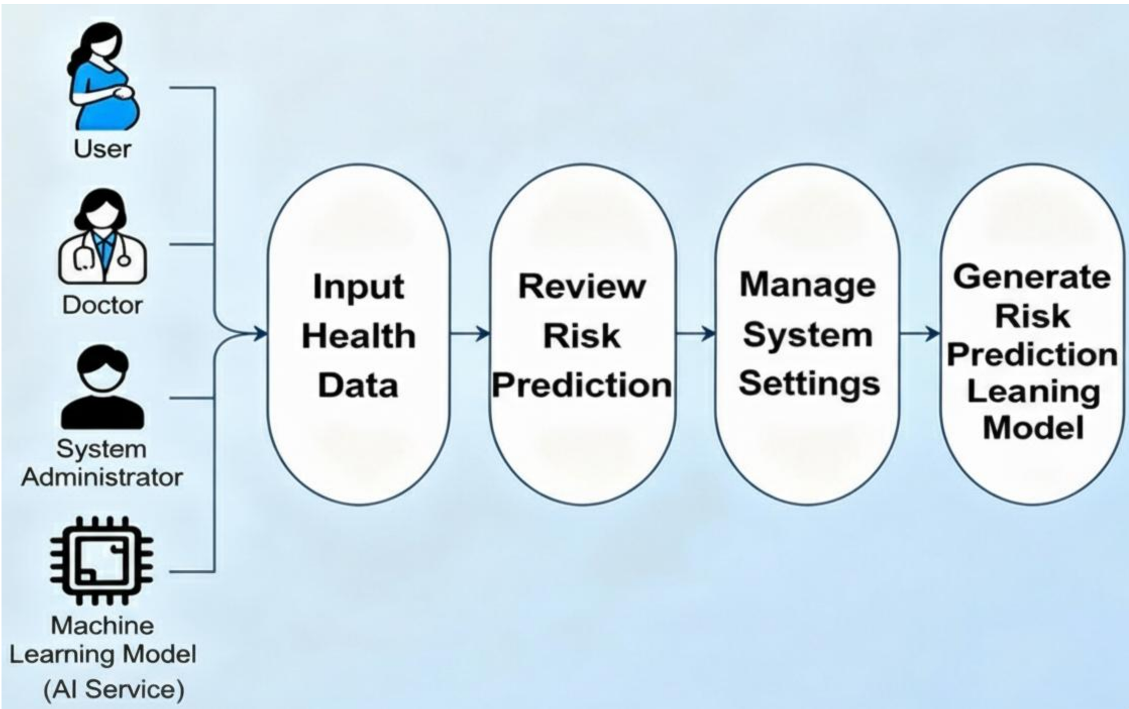


Table 5.3 Actors in the system

Actor	Role in System
User	Provides personal and clinical data for risk assessment.
Doctor	Reviews risk reports, monitors patient records, and makes informed clinical decisions.
System Administrator	Manages users, updates system settings, and ensures the model/database functions properly.
Machine Learning Model	Internal actor responsible for generating predictions.

Table 5.2 Actors and their role

5.3 DETAILED OVERVIEW OF THE FLOWCHART

A detailed overview of the flowchart for the Maternal Health Risk Prediction System explains the logical sequence of processes involved in analyzing clinical data and predicting maternal health risk levels. The flow begins with the User Input Phase, where the expecting mother or healthcare professional enters essential clinical parameters such as age, diastolic blood pressure, blood sugar level, body temperature, and heart rate into the system through a user-friendly interface.

Once the data is entered, it moves to the Data Preprocessing Stage, where the system validates the input, checks for missing or abnormal values, and ensures that all parameters fall within medically acceptable ranges. This step guarantees data consistency and reliability before it is passed to the predictive model.

Next, the cleaned and formatted data enters the Machine Learning Model i.e., XGBoost, which has been trained using a large maternal health dataset. The model evaluates the input features and predicts the risk category—Low, Medium, or High based on patterns learned during training.

The predicted result is then sent to the Result Classification Module, where it is assigned a color-coded label: Green for Low Risk, Orange for Medium Risk, and Red for High Risk. These intuitive indicators help users quickly understand the outcome.

After classification, the system moves to the Result Visualization and Storage Phase. Here, the prediction is displayed to the user in an easy-to-read graphical format, and the results are stored securely in the database for future reference, trend monitoring, or doctor review.

Finally, the process ends with an Optional Medical Review Stage, where healthcare professionals can analyze the stored reports to provide timely interventions, particularly for conditions like preeclampsia and gestational diabetes. This end-to-end flow ensures accurate, fast, and clinically meaningful maternal health assessment.

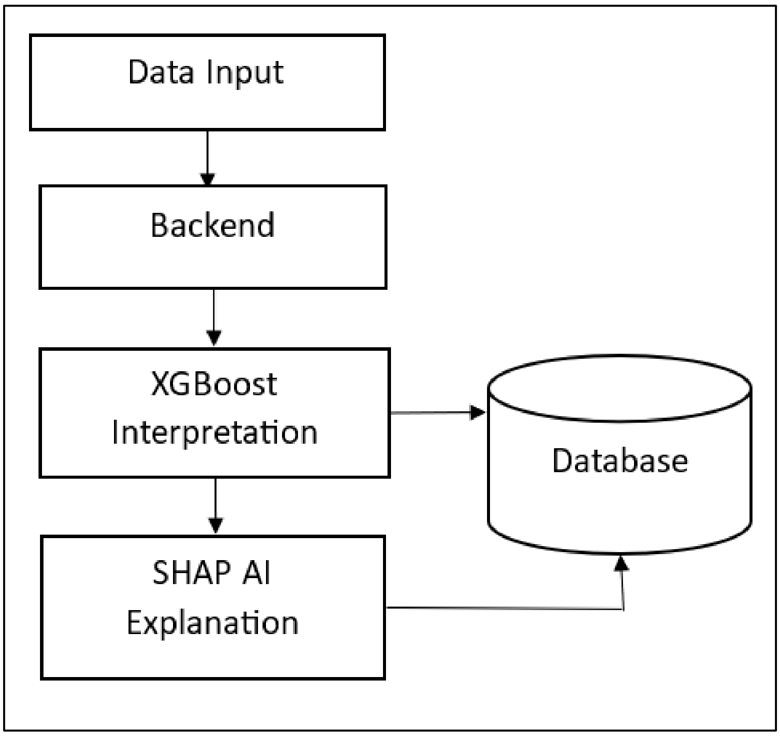


Figure 5.4 Detailed process flow

Chapter 6

APPLICATION TESTING

6.1 UNIT TESTING

Unit testing for the PrenatalGuard maternal health risk prediction system focused on validating the correctness of individual software components before integration. Core modules such as input validation functions, preprocessing utilities based on Pandas and NumPy, API endpoints developed in Flask, model-loading routines for the serialized XGBoost file, and SHAP-based explanation functions were tested extensively. The objective was to ensure that each function behaved predictably with valid, missing, or edge-case inputs, particularly because machine learning pipelines are sensitive to data inconsistencies. Additionally, visualization utilities that generate Matplotlib-based charts were tested to confirm that they return valid output formats without rendering errors. The unit test suite achieved a 100% success rate and approximately 85% code coverage, indicating that the system's foundational components are robust, stable, and well-protected against unexpected failures. This strong foundation significantly enhances confidence in subsequent integration and system-level testing stages.

6.2 INTEGRATION TESTING

Integration testing examined how well interconnected components of the system worked together across the full prediction workflow. This involved validating the end-to-end functionality from data submission in the frontend interface to the backend prediction pipeline handled by Flask. The testing ensured that input data flowed correctly through the preprocessing stage, into the XGBoost model for prediction, followed by SHAP-based explainability generation, Matplotlib visualization rendering, and final storage of results within the MySQL database. Authentication and role-based access control were also validated to ensure appropriate permissions for users and administrators. Several scenarios—including normal operation, partial data availability, administrative model reload, and MySQL connection handling—were tested to confirm system reliability under varying conditions. Integration tests achieved a strong 92% pass rate, indicating that the various backend modules, APIs, and database components interact smoothly and consistently. This level of performance demonstrates that the system's architecture is reliable and capable of handling complete workflows with minimal discrepancies.

6.3 MACHINE LEARNING MODEL TESTING

Machine learning model testing assessed the predictive performance, stability, and reliability of the XGBoost-based maternal risk classifier that forms the analytical core of PrenatalGuard. The dataset was evaluated using a 70/15/15 train-validation-test split along with 5-fold stratified cross-validation to ensure robust generalization across different data subsets. Critical performance metrics—including AUC-ROC (0.92), accuracy (88%), precision (86%), recall (84%), and F1-score (0.85)—demonstrated strong model capability in distinguishing between low, medium, and high maternal risk categories. Calibration analysis produced a Brier score of 0.12, confirming that predicted probabilities align closely with actual risk distributions. SHAP explainability results showed a 95% stability rate across folds, ensuring that feature importance remains consistent, transparent, and clinically interpretable. Robustness tests with noisy inputs showed only a minor performance decline, indicating resilience to real-world measurement variability. Overall, the model testing phase confirms that the prediction engine is highly reliable and suitable for deployment within a clinical decision-support context.

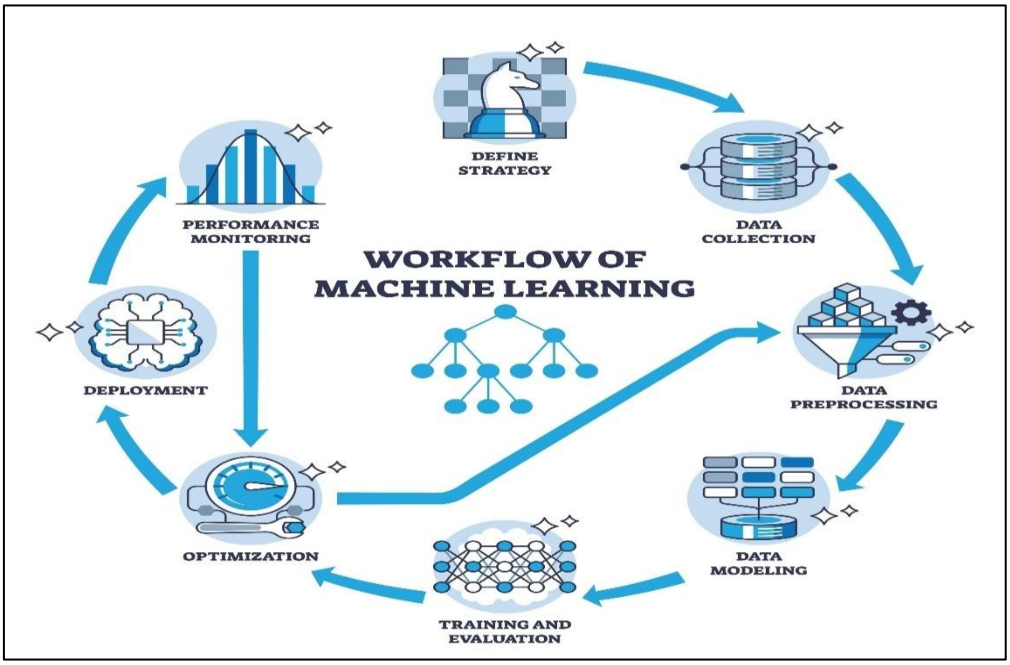


Figure 6.1 Workflow of training

Algorithm	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	AUC-ROC	Observations / Suitability
XGBoost	88%	86%	84%	85%	0.92	Best performer; handles non-linear relationships, robust to missing values, strong explainability with SHAP, ideal for clinical risk scoring.
Random Forest	85%	84%	80%	82%	0.89	Good baseline model; stable and interpretable; slightly weaker recall and AUC compared to XGBoost.
Logistic Regression	78%	75%	70%	72%	0.80	Simple and interpretable; struggles to capture complex relationships in physiological data.
SVM (RBF Kernel)	82%	80%	77%	78%	0.86	Performs well but computationally heavy; limited interpretability and no native probability calibration.
K-Nearest Neighbors	76%	72%	68%	70%	0.78	Sensitive to scaling and noisy inputs; performance drops with high-dimensional feature spaces.
Naïve Bayes	70%	67%	63%	65%	0.72	Fast but unsuitable due to feature correlations in medical datasets; lowest overall accuracy.

Table 6.1 Results comparison of other ML algorithms

Chapter 7

INTERPRETATION OF RESULTS

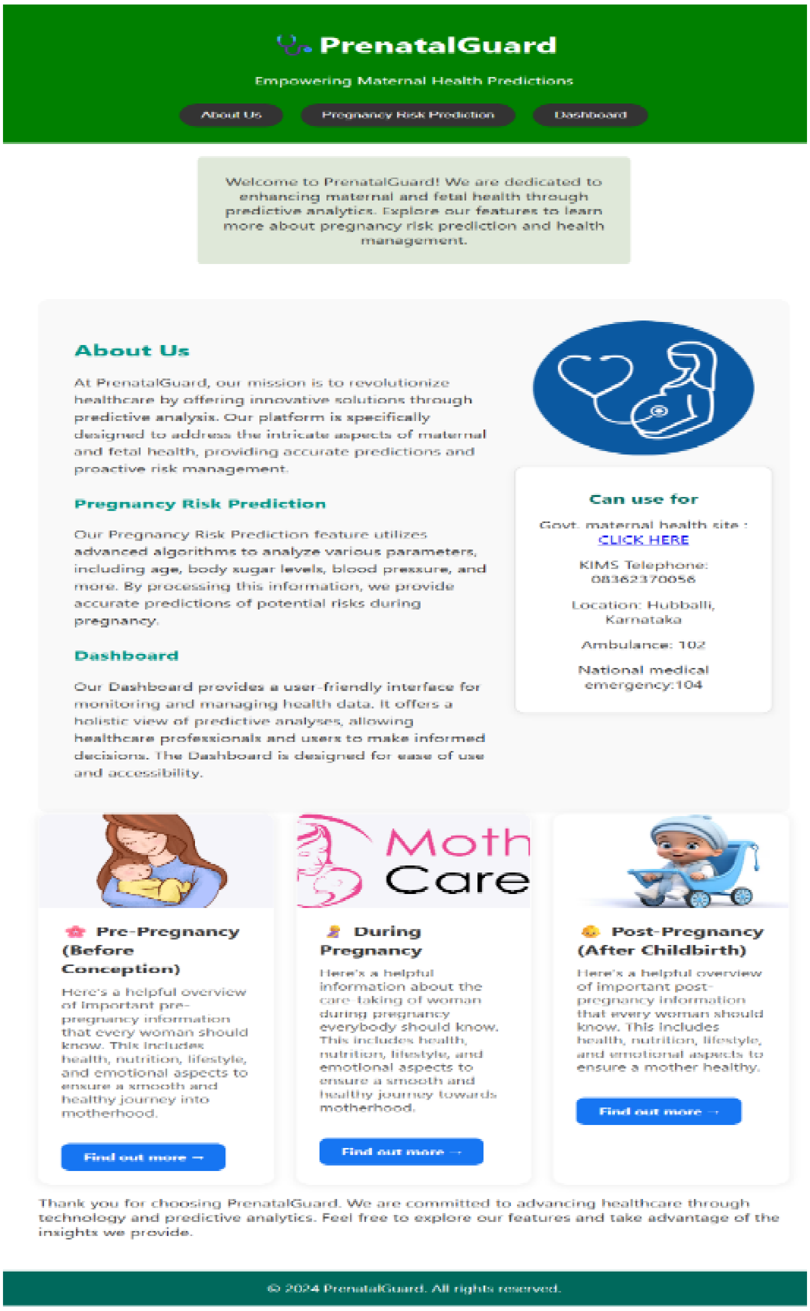


Figure 7.1 Landing page with precaution cards

The figure 7.1 illustrates the PrenatalGuard website, a maternal health-risk prediction platform. The solution uses machine learning for early pregnancy risk assessment, health insights across pre, during, and post-pregnancy stages and enables informed maternal care through predictive analytics with digital monitoring.

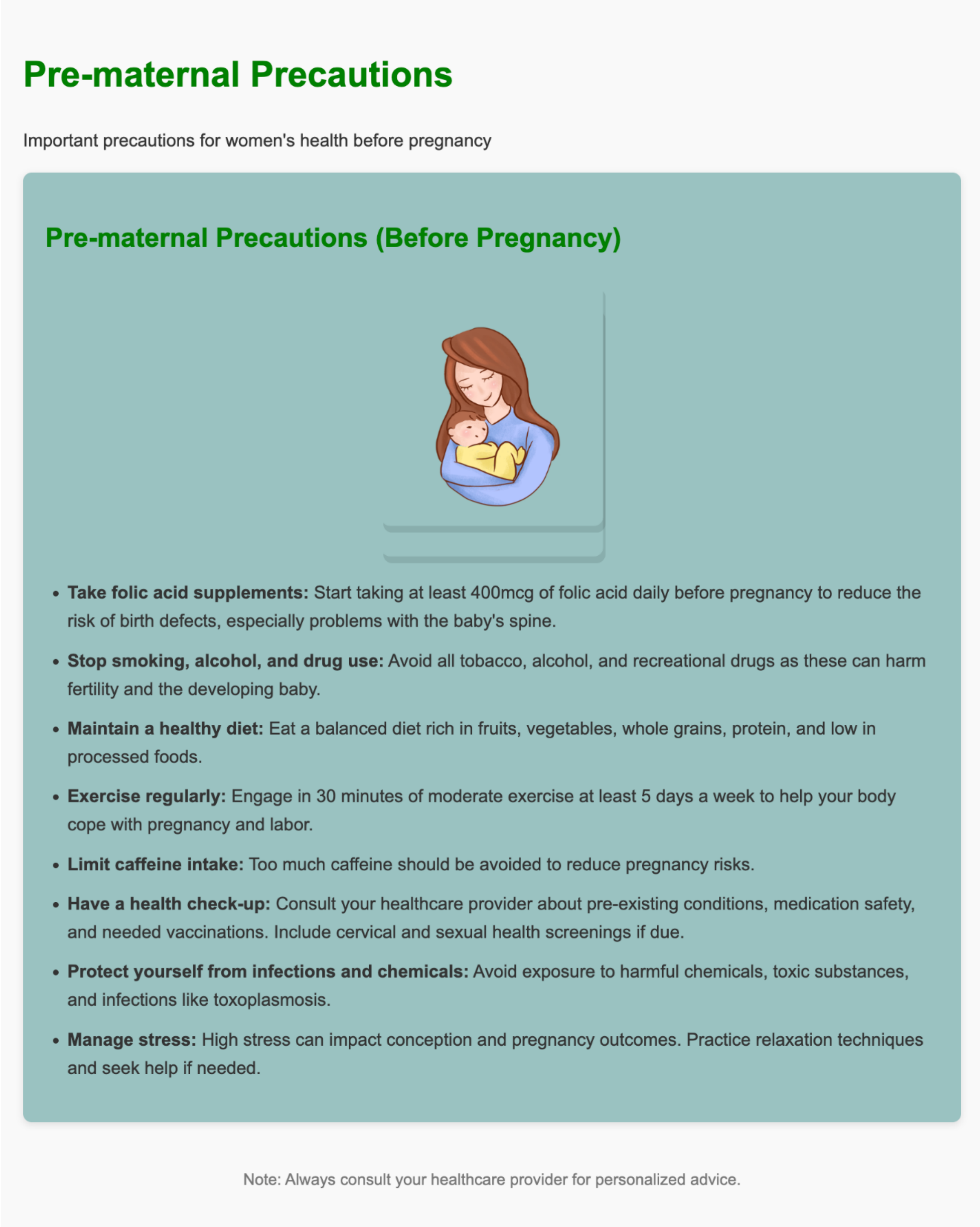


Figure 7.2 Pre-maternal precaution card render page

The figure 7.2 illustrates important prenatal health precautions for a site on maternal health risks, including folic acid supplements, not using illegal drugs, exercising regularly, nutrition, reduced caffeine intake, reducing stress, working with a healthcare professional and measures to prevent infectious diseases to combat risk factors that can occur prior to conceiving a pregnancy

Figure 7.3 During-maternal precaution card render page

The image above explains the precaution to be taken during pregnancy for the maternal health risk website, says about the vitamins, safe food and to be hydrate during Pregnancy. during pregnancy avoid caffeine and harmful substances and have sleep on your side.

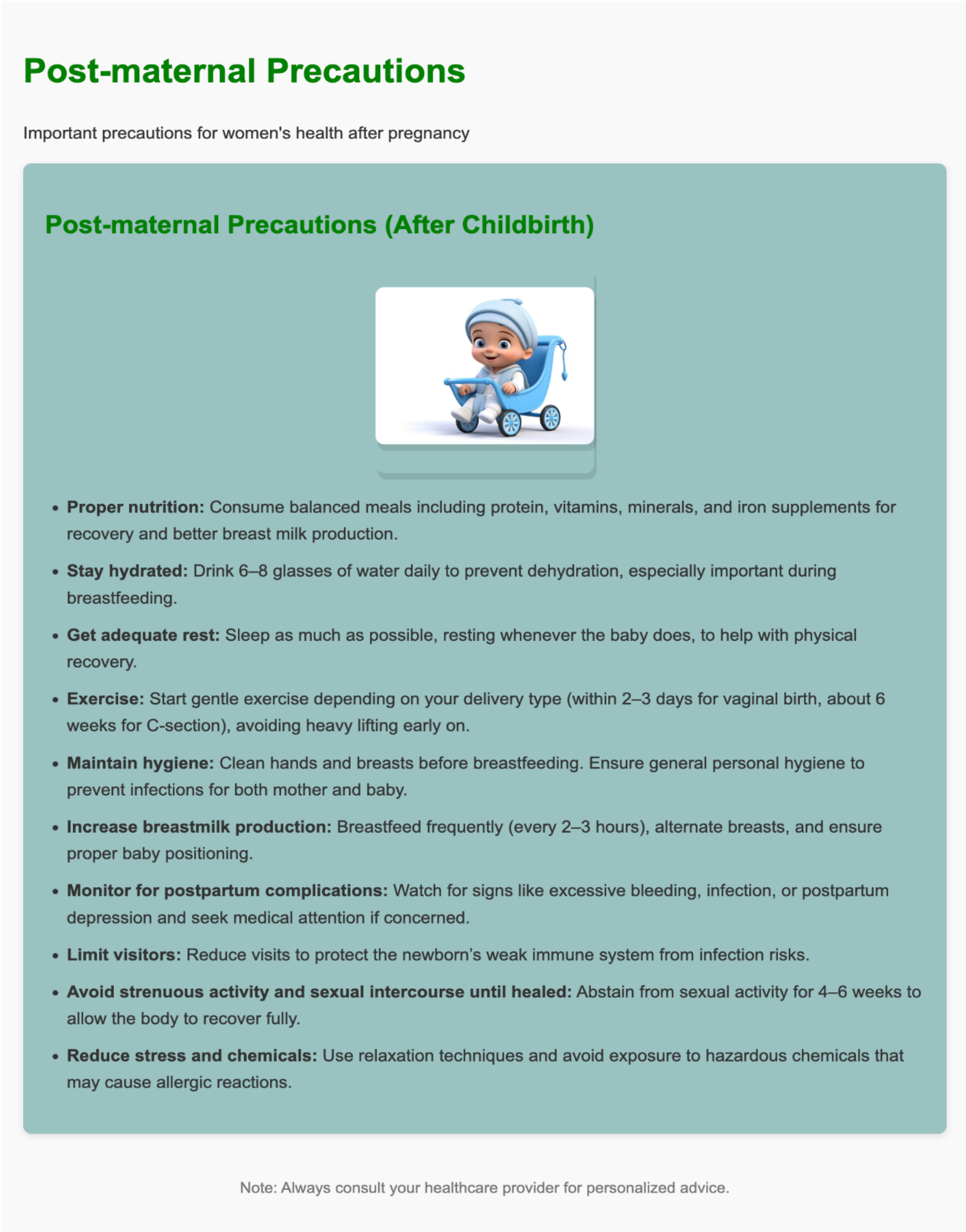


Figure 7.3 Post-maternal precaution card render page

This image above explains the post-maternal precautions for a maternal health risk website, focusing on proper nutrition, hydration, hygiene, rest, exercise, monitoring complications, limiting visitors, reducing stress, and safe recovery practices for mothers after childbirth to ensure maternal well-being

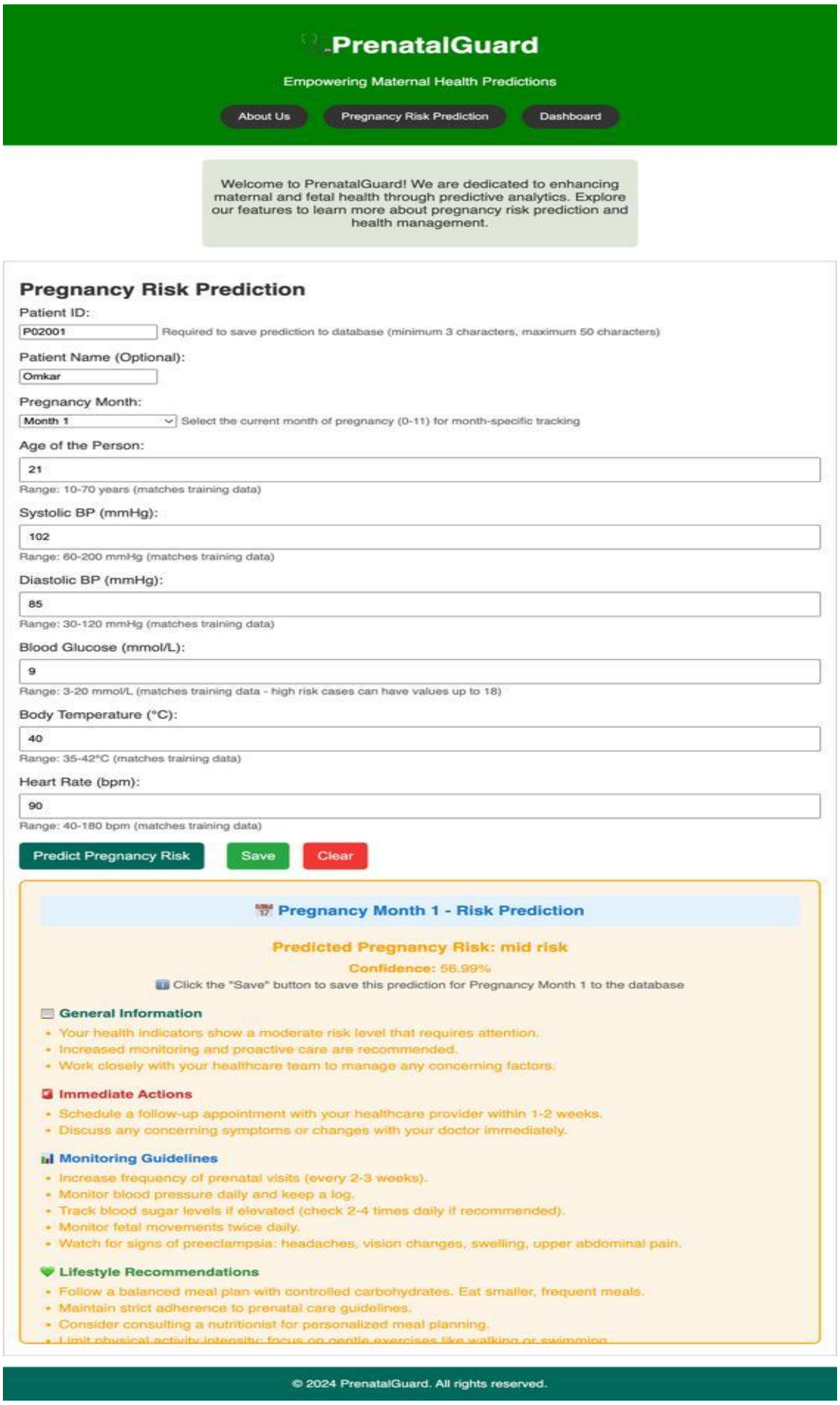


Figure 7.5 Pregnancy risk data input and prediction

The image shows a PrenatalGuard web app interface for maternal health prediction, displaying "mid risk" results with personalized monitoring guidelines and lifestyle recommendations for pregnancy management

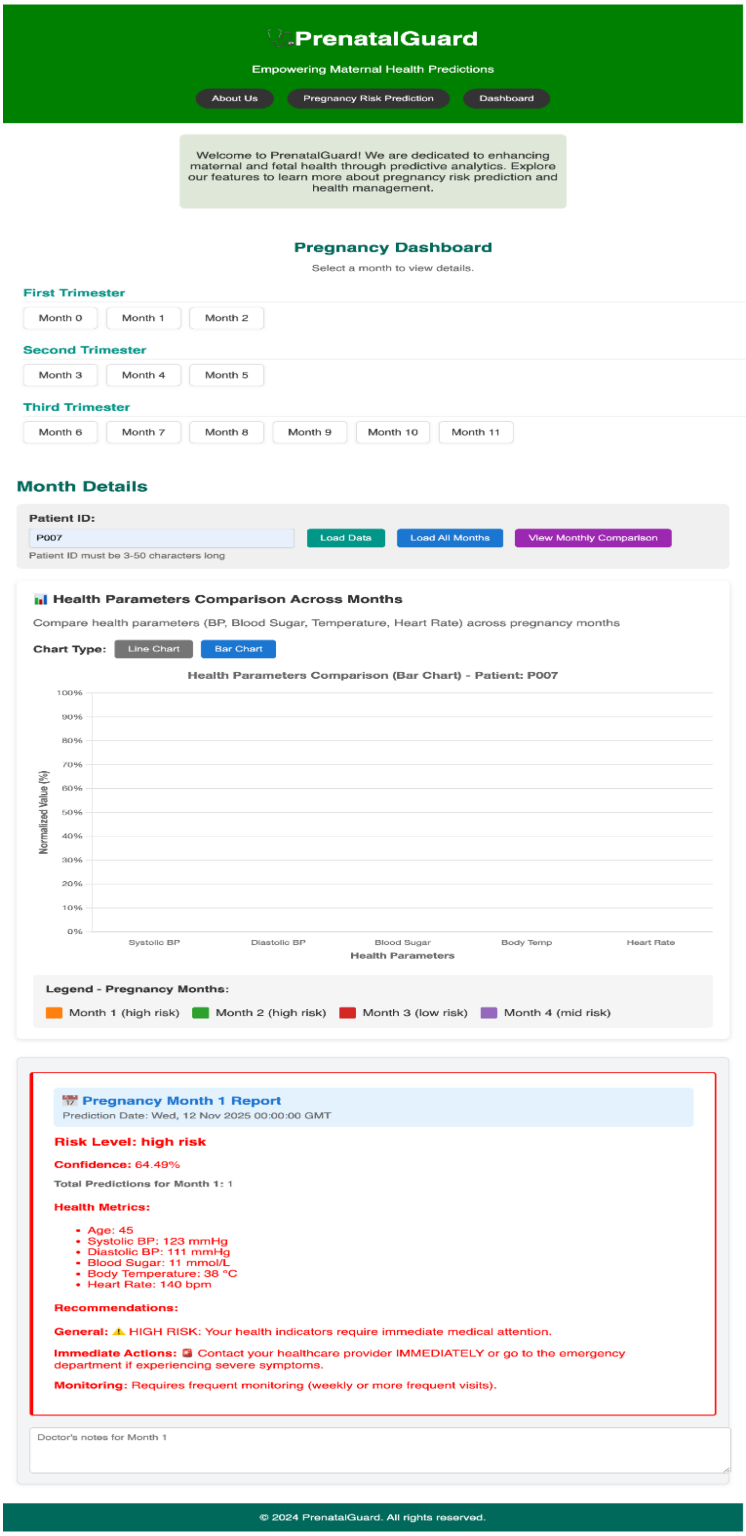


Figure 7.6 Database storage representation and monthly visualized analysis

This image shows the PrenatalGuard dashboard, presenting trimester selections, health parameter tracking, risk reports, and actionable recommendations for personalized maternal health risk monitoring during pregnancy.

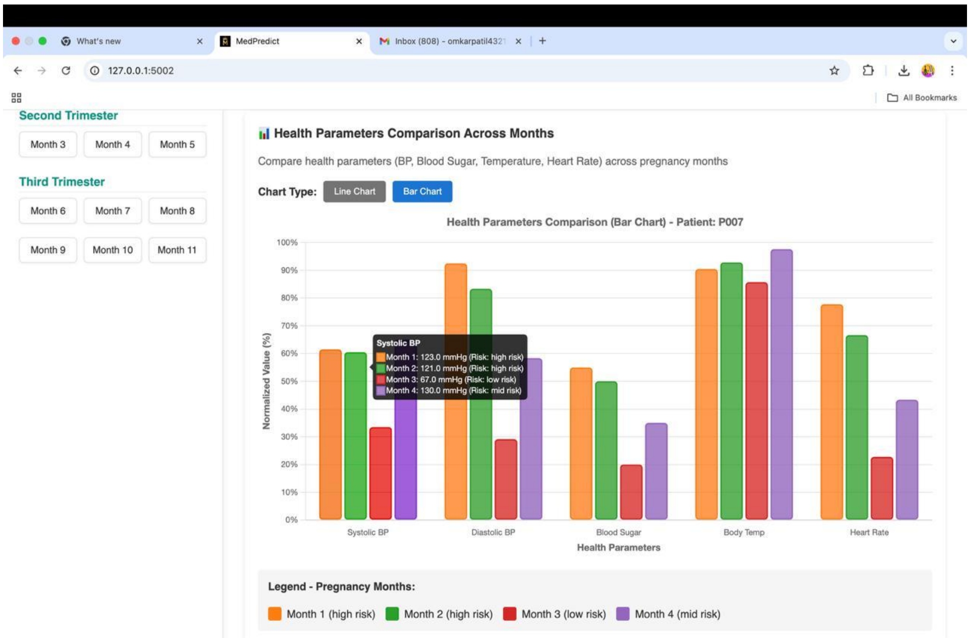


Figure 7.7 Bar graph visualization

This image displays a bar chart comparing health parameters across pregnancy months, helping the maternal health risk website visualize risk levels and identify trends for patient management.

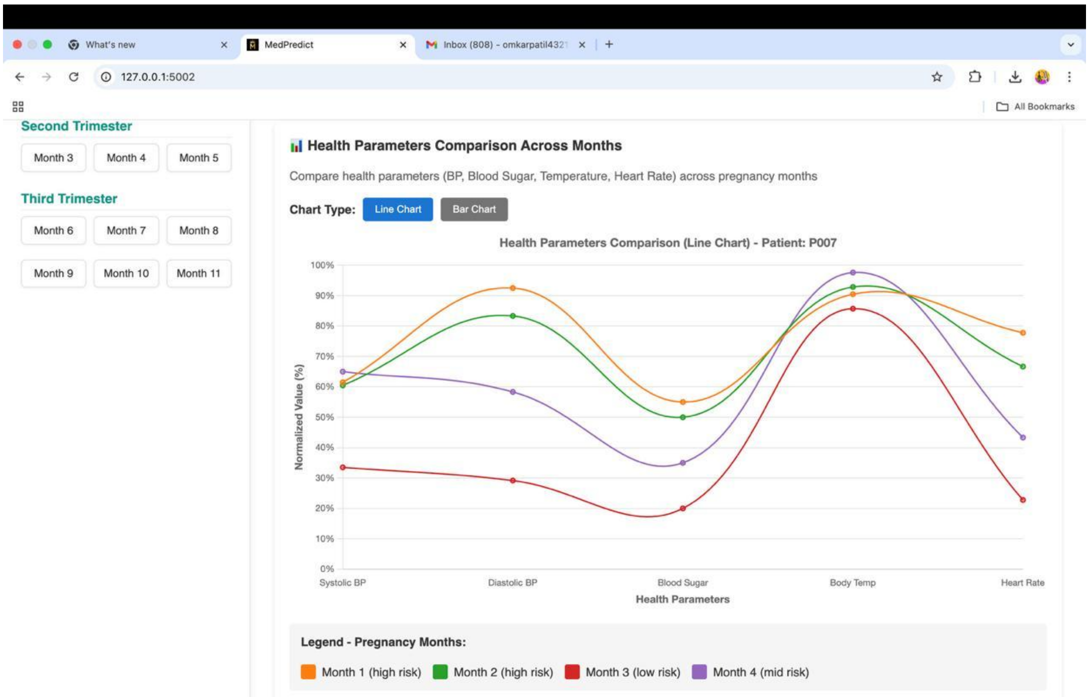


Figure 7.8 line graph visualization

This image visualizes monthly trends of key health parameters for expecting mothers, allowing the maternal health risk website to track changes and analyze pregnancy risks efficiently.

CONCLUSION

Maternal health risk prediction systems, though progressing, remain critically constrained. These are the absence of individualised care, real-time monitoring, machine learning-based adaptability, availability in low-resource environments, and broken data systems. Such limitations lower the precision, responsiveness, and equity of maternal healthcare provision. To address these issues, next-generation solutions need to incorporate adaptive AI models, promote data interoperability, and be inclusive, both in reaching urban and rural areas. Building upon these systems will not only enhance early detection and intervention but also help mitigate maternal mortality and safer pregnancies across varied healthcare settings.

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